

COVID-19 Vaccination Analysis

Yash Anand Rajpal, Advait Bhargav, Glit Hanpanitkitkan

Date

Abstract

Write your abstract here.

Keywords

Vaccine Efficacy, Bayesian Inference, Likelihood Inference, Likelihood Ratio Test

Introduction / Background

The rise of COVID-19 in 2019 resulted in a global crisis affecting healthcare, economics, and daily life across societies worldwide, forcing governments to implement unprecedented public health measures. By the end of 2020, countries such as the United States had seen over a million cases of COVID. This meant the race to find a reliable vaccine was on and it became a primary goal for pharmaceuticals and governments worldwide in order to control the pandemic. Pfizer and BioNTech developed BNT162b2, an mRNA based vaccine, while Janssen and AstraZeneca did not.

The clinical trial was a placebo-controlled, observer-blinded study that enrolled 43,548 participants aged 16 year or older. The participants were randomly assigned in a 1:1 ratio to receive either two doses of the BNT162b2 vaccine or a placebo, both administered 21 days apart. The primary endpoint was laboratory-confirmed COVID-19 occurring at least 7 days after the second dose (Polack et al., 2020).

The results from the trial are stated in the table below:

Group	Cases	Sample_size
BNT162b2	8	17,411
Placebo	162	17,511
Total	170	34,922

Table 1: Vaccine Efficacy Against COVID-19 at Least 7 Days After Second Dose

Statistical Methods

Model

Let X denote the number of COVID-19 cases in the vaccine group among the $n_1 = 17,411$ vaccinated with BNT162b2 participants, and let Y denote the number of cases in the placebo group

among the $n_2 = 17,511$ placebo participants. We assume

$$X \sim \text{Binom}(n_1, \pi_v) \text{ independently of } Y \sim \text{Binom}(n_2, \pi_p)$$

where π_v and π_p are the true infection probabilities in the vaccine and placebo groups, respectfully.

The parameter of interest is the vaccine efficacy:

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{\pi_p - \pi_v}{\pi_p}$$

Rather than working with two binomial random variables, we adopt a single binomial model.

Let's look at an alternative model that we should use instead for this scenario. Let T denote the number of vaccine-group cases among the $n = 170$ total confirmed COVID-19 cases, conditional on $X + Y = 170$. The distribution of T be quite approximated by a binomial distribution.

Then

$$T \sim \text{Binom}(n = 170, \pi)$$

where $\pi = P(\text{aspirin}|\text{case})$ and can be shown to be equal to

$$\pi = \frac{n_1 \pi_v}{n_1 \pi_v + n_2 \pi_p} = \frac{\pi_v}{\pi_v + \pi_p}$$

The vaccine efficacy ψ can be written in terms of π as:

$$\psi = \frac{1 - 2\pi}{1 - \pi}, \quad -\infty < \psi < 1$$

We can interpret ψ understanding that:

1. when $\pi = 0$ (no vaccine cases), $\psi = 1$ (perfect efficacy)
2. when $\pi = 0.5$ (equal rates), $\psi = 0$ (no efficacy)
3. $\psi < 0$ means that the vaccine is harmful and should definitely not be used.

From the table, we observe $t = 8$ vaccine cases among the $n = 170$ total cases. This seems very small, but we should still conduct tests to verify.

We are interested in an interval estimate for this parameter (ψ).

The hypothesis test is: $H_0 : \psi = 0.30$ versus $H_1 : \psi > 0.30$ since the FDA requires at least 30% efficacy for a new therapy to be approved.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

In a Bayesian framework, we specify a

Results

Present your findings.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example.com>.

Appendix

Code

```
# You can reference your code in the appendix (sample here).
```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.